

A Persona Changes How a Decision Model Trades, Not How Well

Jev as a market participant in the card game Figgie

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<https://github.com/jaceyang97/figgie-on-jev>

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Abstract

Can we tell an AI model which kind of trader to be, and then use it as that trader in a simulated market? We test this with Jev, a model that reads a situation and gives a probability for each option in a list. For each of six rule-based traders with a published algorithm, we write the algorithm as a text *persona* and give it to Jev. Then we put Jev in the trader’s seat, with the same cards and the same opponents, and compare the two. The market is Figgie, a card game that copies a trading floor, under two sets of market rules. At 1,200 frozen game moments, the persona moves Jev’s actions toward the rule-based trader in 22 of 24 cases, but the two still agree on at most 57% of their probability. Jev’s belief about which suit pays is worse than a uniform guess, also when the facts decide the answer. In 1,040 paired games, Jev earns less than the trader it plays in both rule sets (−33 and −58 chips per game), and it is never clearly better. A persona controls Jev’s style of trading. It does not give Jev the judgement that the algorithm contains.

1 Introduction

Look at one moment of one game (Figure 1). The game has run for 207 of its 240 seconds. The player’s hand and the public record of trades show that at least 11 different clubs exist. In Figgie, only one suit has more than 10 cards, so clubs is the 12-card suit. The rules say that the suit that pays, the *goal suit*, has the same colour as the 12-card suit. So the goal suit is spades. The player holds 6 spades. A careful player now knows the answer to the only question that matters in the game.

We asked Jev, a decision model, the same question at this moment. Jev reads a description of a situation and returns a probability for each option in a list [3]. It gave spades a probability of 0.48 and clubs 0.41. When we added the complete list of trades to its state, it gave clubs 0.45. When we added the count of known cards, which states that 11 clubs exist, it gave clubs 0.59. Then we told Jev to trade as the *fundamentalist* of Ozerov, DiSilvio and Luo [1], and we gave it the fundamentalist’s full algorithm as text. The rule-based fundamentalist at this moment mostly passes (probability 0.63). Otherwise it sells a club, which pays nothing, bids a few chips for a spade, or offers a spade at 32 chips or more. Its formula values the sixth spade at about 32 chips, because a player that sells it may lose the majority prize. Jev, with the same formula in its instructions, sold a spade at the standing bid of 12 chips with probability 0.18.

We chose this moment because it is the clearest case in our data. This paper asks whether what happens in it is typical. The question behind it is practical. People now propose AI models as stand-ins for people in economic experiments [4], and as traders in simulated markets. A model that reads text can be told who to be. If this works, a market simulation can get a crowd of different participants from a few sentences each. If it does not work, the simulation contains traders that look different on paper but behave in ways that nobody chose.

To answer the question, we need traders whose behaviour is known exactly. The Figgie study of Ozerov et al. gives four such traders with complete algorithms, and the market-making literature gives two more [6, 7]. Each rule-based trader is Jev’s *twin*: we give Jev a text version of the twin’s algorithm, which we call a *persona*, put Jev in the twin’s seat, deal the same cards, and let the same opponents act at the same times. Any difference between Jev and its twin then comes from Jev.

Our central claim is this: *a persona changes how Jev trades, but not how well*. We support it at three levels, and each level uses its own data.

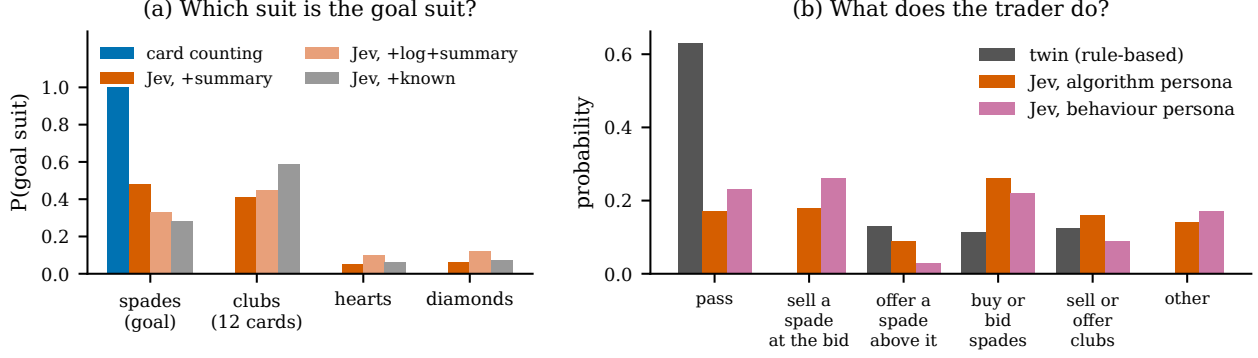


Figure 1: At one moment where the facts decide the goal suit, Jev does not find it, and a persona does not make it act like the trader it describes. Rule set A, game 24, seat 0, 207 s into the game. (a) The public trades show 11 clubs, so clubs is the 12-card suit and spades is the goal suit; card counting (the exact calculation from the known cards, section 2) gives spades probability 1. Jev gives clubs, a suit that cannot pay, 0.41 to 0.59, even when the full trade list (+log+summary) or the count of known cards (+known) is in its state (section 4). (b) The rule-based fundamentalist (the *twin*) mostly passes; otherwise it sells clubs (worth 0), bids low for spades, or offers spades at 32 chips or more. Jev, given the twin’s algorithm as text, sells a spade at the standing bid of 12 chips with probability 0.18.

1. **Belief** (section 7). Jev’s estimate of the goal suit is worse than a uniform guess over the four suits. It does not improve when we give it more facts, and it does not improve during a game.
2. **Decisions** (section 8). A persona moves Jev’s actions toward the twin in 22 of 24 cases. But Jev follows the words of a persona, not its procedure, and the similarity stays low.
3. **Profit** (section 9). In 1,040 paired games, Jev earns less than its twin in both rule sets, and no persona makes it clearly better than its twin.

Before these results, section 6 shows that our simulator reproduces the published behaviour of the rule-based traders. Section 10 answers the objections that we expect from a careful reader.

2 Figgie, and what a trader must know

Jane Street designed Figgie to teach trading [2]. It is small enough to simulate exactly and rich enough to need a view of value.

Rules. The deck has 40 cards in four suits: one suit has 12 cards, one has 8 and two have 10. Nobody knows which suit has how many. The goal suit is the suit of the same colour as the 12-card suit, so it has 8 or 10 cards. Each of four players starts with 350 chips, pays 50 into a pot of 200, and gets 10 cards. For 240 seconds, the players buy and sell single cards for chips. At the end, each goal-suit card pays 10 chips from the pot, and the player with the most goal-suit cards gets the rest (100 chips if the goal suit has 10 cards, 120 if it has 8; a tie splits it). A player’s *profit* is its final chips minus 350. The game is zero sum.

What the facts say. There are 12 possible decks: 4 choices for the 12-card suit times 3 choices for the 8-card suit. Before any trade, each deck is equally likely. A player learns about the deck from two sources: its own hand, and the cards that other players reveal when they sell. The paper’s Algorithm 3 counts, for each suit s , the number k_s of different cards known to exist (a card counts once, even if it changes hands many times). A deck in which suit s has n_s cards can produce this evidence in $\binom{n_s}{k_s}$ ways, so

$$P(\text{deck} \mid k) \propto \prod_s \binom{n_s(\text{deck})}{k_s}, \quad P(\text{goal} = s \mid k) = \sum_{\text{decks with goal } s} P(\text{deck} \mid k).$$

We call this calculation *card counting*. In Figure 1, $k_{\text{clubs}} = 11$, so only decks with 12 clubs remain, and all three of them have spades as the goal suit. Card counting is simple, and it uses only facts that every player

sees. It is our reference for belief. It is not a strong reference: at the frozen moments of section 7, its score is close to that of a uniform guess, probably because most moments reveal few cards.

3 The traders and the market

Six rule-based traders. The four traders of Ozerov et al. send orders with one common rule (their Algorithm 2). A trader picks a suit, then buy or sell with equal chance. To buy, it draws a price between 0 and its buy value; if the best offer is at or below that price, it takes the offer, otherwise it bids the price. To sell, it draws a price between its sell value and twice that value, and acts in the same way. The four traders differ only in how they value a suit.

- The **fundamentalist** values a suit from card counting. Its buy value for one more card of suit j , when it holds n , is $e_b(j, n) = \sum_i P(i | k) v(i, j, n)$ over the 12 decks i . Here $v = 0$ if j is not the goal suit of deck i ; otherwise $v = 10 + a_i r^n$ while n is below the majority size x_i (5 of 8, or 6 of 10), and $v = 10$ after that. The term $a_i r^n$ with $r = 1.2$ and $a_i = p_i(1 - r)/(1 - r^{x_i})$ shares the majority prize p_i over the cards needed to win it. The sell value is $e_s(j, n) = e_b(j, n - 1)$. The fundamentalist also cancels its orders that no longer agree with its values (the paper’s Algorithm 4).
- The **bottom-feeder** knows nothing about the deck. It copies the fundamentalist: its value is the middle of the fundamentalist’s last 4 buy prices and last 4 sell prices.
- The **chartist** expects the recent trend of prices to continue [5].
- The **noise trader** values a suit at the best bid times e^Z , with $Z \sim N(0, 1)$. It does not trade a suit that has no bid.

We add two market makers. They quote both sides and never take another trader’s order. The **Glosten–Milgrom** market maker (GM) [6] starts from card counting and updates its belief with the direction of each trade, assuming that a share $\mu = 0.3$ of the traders are informed; it asks the expected card value after a buy and bids the value after a sell. The **Avellaneda–Stoikov** market maker (AS) [7] quotes around the market price, moves its quotes against the cards it holds, and quotes wider when prices move a lot and early in the game (risk aversion $\gamma = 0.1$, order-arrival decay $k = 1.5$).

Two market rules. Ozerov et al. use an order book, while the real game uses open outcry. The two rules reward different skills, so we use both.

- **Rule set A, order book** (the paper). Many orders can wait at each price. A new order trades with the best waiting order on the other side, at the waiting order’s price, oldest first. A player keeps at most 5 orders per side and suit and can cancel them.
- **Rule set B, open outcry** (the real game). Each suit has one best bid and one best ask. A new bid must be higher than the best bid, and a new ask must be lower than the best ask. Every trade removes all bids and asks.

Time. The simulator is event driven, as in the paper. Each trader wakes after a random gap with mean $1/\lambda$, looks at the market and decides. An order reaches the market Λ seconds after the trader looked. In all tests of Jev, every trader, Jev included, wakes $\lambda = 0.5$ times per second and has a delay of $\Lambda = 0.3$ seconds. So a difference in profit cannot come from speed. For the replication, we also use the paper’s speed (one wake per second, no delay) and the paper’s end rule (10,000 events instead of 240 seconds).

4 Jev as a trader

A Jev trader works like the rule-based traders: it wakes, looks at the market and acts. Only the decision is different. Jev (`typesafe/jev-1.13`) reads a JSON *state* of up to 64,000 tokens and answers *choice* questions. A choice question has up to 255 options, each with a one-line description, and Jev returns a probability for every option. We call Jev through OpenRouter; input costs US\$0.042 per million tokens.

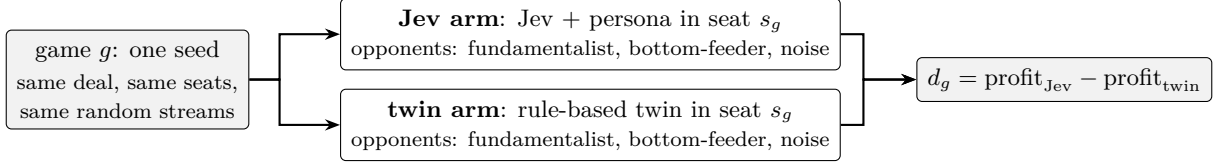


Figure 2: The twin test changes only the trader in one seat. Both arms of game g use the same seed. The tested seat rotates from game to game. At frozen moments of the twin arm, we also ask Jev what it would do, and compare its probabilities with the twin’s (section 8).

Each time the Jev trader wakes, it sends one request with two questions: “which action do you take now?” and “which suit is the goal suit?”. The request has four parts.

1. **Rules.** A short, correct text of the rules of Figgie and of the rule set. It contains no strategy hints and no conclusions drawn from the rules. For example, it states the colour rule, but not that many clubs imply a spade goal.
2. **State.** Jev’s hand, chips, time left and the order book, plus one of five versions of context. Three versions contain only facts: *basic* (nothing more), *+summary* (per-suit trade counts and last prices, each player’s net buys and sells, and Jev’s own recent actions and their results) and *+log+summary* (also the full list of trades). Two versions add numbers that code calculates, and we use them only as tests: *+known* (the k_s of card counting) and *+assist* (the card-counting probabilities themselves).
3. **Persona.** None (*neutral*), or one of two wordings for each of the six twins. The *algorithm* wording states the twin’s steps and numbers; for the fundamentalist it is the formula above, in words. The *behaviour* wording describes what the trader does, without steps or numbers: “You value each suit only from the cards you know exist ... You bid below your value and offer above it, choosing buying or selling and the price with some randomness.”
4. **Actions.** Pass; take the best bid or offer; cancel (rule set A); or bid or offer at one of 30 prices (1 to 20, then 22 to 40 in steps of 2). Only actions that the rules accept are listed, so there are at most 253 options.

Jev’s action is a random draw from its probabilities. We call this *sampling*. An earlier version of this study took Jev’s most probable option instead. Jev then passed at most of its turns, because its probability of acting was split over many prices, so the single pass option was usually the most probable one. Sampling also matches the twins, whose algorithms are random by design.

5 The twin test

The idea is simple: change only the trader in one seat, and keep everything else fixed (Figure 2). Game g has a seed. The seed fixes the deal, the seats, and one random stream per seat for wake-up times and random choices. So in game g , the Jev arm and the twin arm have the same cards in every hand, and the three opponents draw the same random numbers. The only thing that differs is the trader in the tested seat. The opponents still react to what that trader does, which is the point: we measure the trader in a live market.

We run the study in three stages.

Stage 0: are the twins right? Before we compare Jev with a twin, the twin must behave as published. We replay the line-ups of the paper’s figures 3 to 5 and tables 2 and 3, 100 games each, with rule-based traders only. We use three settings (A with 10,000 events, A with 240 seconds, B with 240 seconds) and both speeds. This stage costs nothing.

Stage 1: decisions at frozen moments. We play rule-based games in the line-up of stage 2 and freeze the tested seat’s view at random wake-ups where the twin can act: 100 moments for each rule set and twin type, at most 3 per game, 1,200 moments in total. At each moment we know the true goal suit, the card-counting

probabilities, and the twin’s probability for each action (from 4,000 draws of its random choices). We then ask Jev.

We compare two probability distributions over the same menu, Jev’s π and the twin’s τ , by their *overlap*,

$$\text{overlap}(\pi, \tau) = \sum_a \min(\pi(a), \tau(a)) = 1 - \text{TV}(\pi, \tau),$$

the share of probability that the two place on the same actions. It is 1 for identical behaviour and 0 for no common action. A twin price that is not on Jev’s price list moves to the nearest listed price (3.1% of the twin’s probability). We judge a belief p over the four suits by its *log loss* $-\ln p(\text{goal})$, which punishes confident mistakes most; a uniform guess scores $\ln 4 \approx 1.386$. We also report the Brier score $\sum_s (p_s - \mathbf{1}[s = \text{goal}])^2$ (uniform: 0.75). Stage 1 has three parts: *1a* asks each moment with the algorithm persona, the behaviour persona and no persona (3,600 requests); *1b* asks the fundamentalist moments with no persona in all five state versions (1,000 requests); *1c* asks 50 moments twice (100 requests). Stage 1 cost US\$1.02.

Stage 2: profit in full games. Jev plays against a fundamentalist, a bottom-feeder and a noise trader, the trader types of the paper’s line-ups. (We leave out the chartist as an opponent, because it loses so much that it gives free profit to the others.) There are 26 conditions: 2 rule sets times (neutral plus 6 twins times 2 wordings). The twin of neutral Jev is the fundamentalist, the only rule-based trader that makes money. Each condition plays 40 games, a number fixed before the run, and every condition uses the same 40 seeds. Our measure is the paired difference d_g of Figure 2. Stage 2 used 117,789 requests and cost US\$29.98.

Uncertainty. All intervals are 95% intervals from a bootstrap that resamples whole games (4,000 resamples), because all results from one game number share its deal. In stage 1, the 24 comparisons of a persona with no persona use a Holm correction. For stage 2, we fit one regression of d_g on the design,

$$d_g = \beta_0 + \beta_A \mathbf{1}[A] + \beta_D \mathbf{1}[\text{behaviour}] + \beta_{AD} \mathbf{1}[A] \mathbf{1}[\text{behaviour}] + \sum_k \gamma_k \mathbf{1}[\text{type} = k] + \varepsilon_g,$$

with the same bootstrap (1,000 resamples), and a second time with the number of goal cards in the starting hand added.

What we predicted. We wrote down our predictions before we collected the data (Table 1). We also fixed one rule to choose Jev’s state for stage 2: use the cheapest factual version whose log loss is not clearly worse than the best one.

6 The simulator reproduces the published traders

The twins behave as the paper reports (Figure 3). In both 240-second settings, the fundamentalist earns the most in every line-up, and the paper’s order (fundamentalist, then bottom-feeder, then noise trader) holds in all four line-ups with a bottom-feeder. The noise trader loses in every line-up and every setting. The more fundamentalists a game has, the less each one earns.

The replication is not exact, and we report where it differs. In the paper’s 10,000-event setting, our bottom-feeder beats the fundamentalist in two line-ups (f+b+2n: 360 vs 306 chips; 2f+b+n: 152 vs 96). In that setting our noise traders lose almost all their chips, and our profits are larger than the paper’s; in 240-second games they are smaller. The paper’s values lie between our two settings.

The spread of profits also matches, with one exception. In games of four fundamentalists, the spread of final wealth, cash and payout is 57.8, 30.0 and 67.8 chips (paper: 61.8, 33.1, 68.6; appendix Table 4). Games of four noise traders do not trade at all, while the paper reports trades. The paper does not say what a noise trader does in a suit with no bid; in our reading it has no value there, so a market of noise traders never starts.

Speed acts as in the paper. In the paper’s latency test (its table 3), a slow fundamentalist ends with more cash than a fast one (+47 to +51 chips) and with less payout (−33 to −39); the paper’s signs are the same (appendix Table 5).

Prediction (written before the data)	Result	Main evidence
<i>Belief</i>		
H1a With full facts, Jev’s belief is not worse than card counting	not supported	log loss +0.17 to +0.26 (§7)
H1b With card-counting numbers in its state, Jev is too confident	supported	top probability 0.69, given 0.49
H1c Jev’s belief improves later in the game	not supported	Figure 7
<i>Market rules</i>		
H2a In A, the traders rank as in the paper: fundamentalist, bottom-feeder, noise	partly	yes in 240-s games; not in 10,000-event games
H2b In B, the paper’s traders trade less and the fundamentalist’s edge is smaller	supported	35 vs 114 trades; 112 vs 222 chips
H2c The profit gap between Jev and its twin differs between A and B	supported	$A - B = +25$ [12, 36]
<i>Persona wording</i>		
H3a Jev obeys simple yes-or-no rules with both wordings	partly	“never take” broken 1–14%
H3b Jev does not do calculations that a persona asks for	supported	fundamentalist overlap 0.36–0.43
H3c A behaviour text changes actions in ways it does not state	supported	noise trader buys 57%, twin 8%
H3d The wording effect is the same in A and B	not supported	algorithm better for 5/6 types in A, 2/6 in B
<i>Profit</i>		
H4a Neutral Jev earns less than the fundamentalist, more than the noise trader	partly	less than both in A
H4b The more similar the decisions, the more similar the profit	not supported	$\rho(\text{overlap}, d) = +0.53$

Table 1: Most predictions about Jev failed in the direction of “Jev is worse”. The predictions and the primary measures were fixed before any request to Jev.

Rule set B changes the market as we expected. With one fundamentalist and three noise traders, a game has 35 trades in B and 114 in A, and the fundamentalist’s advantage falls from 222 to 112 chips.

Last, we measured how much profit varies by chance in the stage 2 line-up. The difference between two rule-based types in the same seat and games has a standard deviation of 62 to 97 chips. So 40 paired games can detect a mean difference of about 28 to 43 chips in one condition (appendix Table 6). This sets the resolution of section 9.

7 Jev does not know the goal suit

A fundamentalist is only as good as its belief about the goal suit. So we first ask how good Jev’s belief is (Figure 4; appendix Table 7).

More facts do not help. In every factual state version, Jev’s log loss is higher than that of card counting at the same moments: by 0.17 to 0.26 in A and by 0.22 to 0.25 in B. The paired interval excludes 0 in five of these six cases; the exception is +log+summary in A ($[-0.01, 0.34]$). Jev’s mean log loss is also above that of a uniform guess (1.47 to 1.65, against 1.386). The best version is +log+summary in A (1.47), and even there the full list of trades did not stop Jev from backing a suit that the trades rule out (Figure 1).

Jev is most wrong when it is most sure. When Jev gives a suit a probability between 0.4 and 0.6, that suit is the goal suit only 8 to 15% of the time (Figure 4, right). A uniform guess would be right 25% of the time.

When the facts decide, Jev does not see it. Moments where the facts decide the goal suit, as in Figure 1, are rare: it is the only one among the 200 fundamentalist frozen moments. The full games of stage 2 give many more. There, card counting was certain of the goal suit at 2,848 of Jev’s 115,352 decisions (1,989 in A and 859 in B, in 92 of the 1,040 games). At these decisions, Jev gave the true goal suit a mean probability of 0.26 in A and 0.29 in B, about what a uniform guess gives (0.25). It gave the rest, 0.74 and 0.71, to suits that the public facts had ruled out.

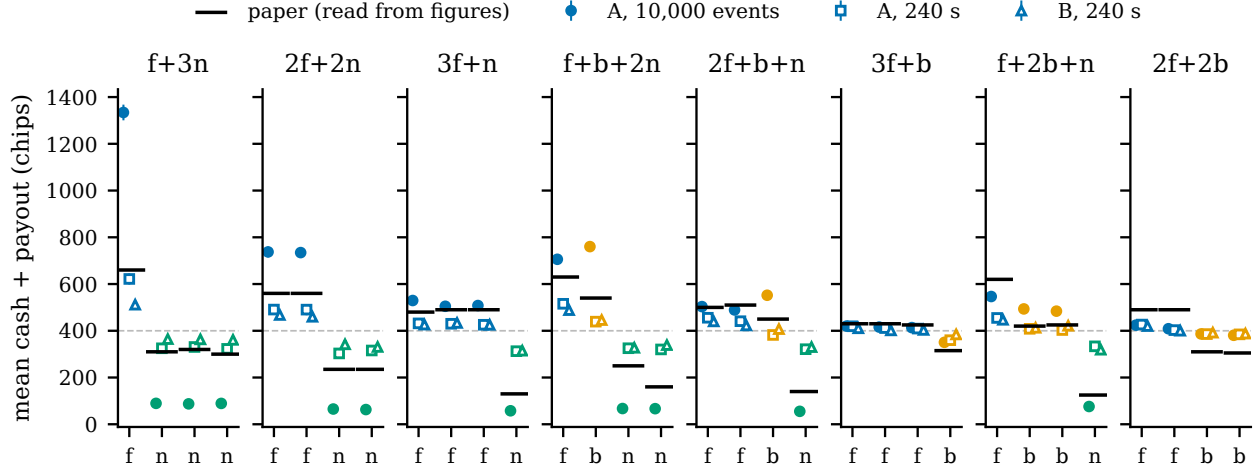


Figure 3: The rule-based traders rank as in the paper: fundamentalist first, noise trader last (in 240-second games; section 6 gives the exceptions in 10,000-event games). Mean final cash plus payout per seat (the paper’s measure; profit + 400), 100 games per line-up, paper’s speed. Black bars: the paper’s values, read from its figures. Error bars: ± 2 standard errors. f = fundamentalist, b = bottom-feeder, n = noise trader. Dashed line: break-even.

The right numbers make it overconfident. With the card-counting probabilities in its state (+assist), Jev finds the most likely suit as often as card counting does (top-1 accuracy 0.45 in A, 0.37 in B). But it makes the numbers more extreme: its mean top probability is 0.69, while the numbers it was given have a mean top probability of 0.49. Its log loss rises to 1.95 in A and 1.80 in B, the worst of all versions.

Later is not better. As a game goes on, more cards are revealed. Jev’s log loss does not follow: with +summary it falls from 1.55 to 1.48 in A and rises from 1.57 to 1.69 in B, from the first to the second half of the moments. Section 9 shows the same in full games (Figure 7).

By our rule, stage 2 uses the +summary state: against the best factual version, it is not clearly worse (A: +0.043, $[-0.006, 0.091]$; B: +0.011, $[-0.037, 0.057]$), while basic is clearly worse in A (+0.092, $[0.007, 0.183]$). A +summary request has about 5,700 tokens in A and 4,900 in B.

8 A persona moves Jev toward its twin, but not far

Figure 5 shows how close Jev’s actions come to the twin’s at the 1,200 frozen moments (appendix Table 8).

A persona moves Jev, and the movement is real. In 22 of 24 cases (6 types, 2 wordings, 2 rule sets), the persona raises the overlap with the twin above that of neutral Jev, after the Holm correction. The gains are 0.01 to 0.21. Two cases move away from the twin: the fundamentalist algorithm in B (-0.035) and the behaviour noise trader in B (-0.063). The effect is not noise from Jev: when we asked the same 50 moments twice, the answers differed by a total variation of only 0.071 for the action and 0.049 for the goal suit, and the most probable action was the same in 96% of pairs.

But the similarity stays low. The highest overlaps are for the noise trader (0.55, algorithm wording, both rule sets) and the bottom-feeder in B (0.57). The fundamentalist, the type that needs the most calculation, reaches only 0.36 to 0.43. Jev does not do the calculation that its persona describes.

Jev follows the words. Jev obeys a short rule better than a procedure, but not always. Both market-maker personas say “never take another trader’s order”; Jev still takes one in 1 to 14% of its probability. Their overlap with the twin is the lowest of all (0.04 to 0.10), because each market-maker twin posts one exact price per side, and Jev spreads its probability over many prices. Words can also act in ways that the text does not

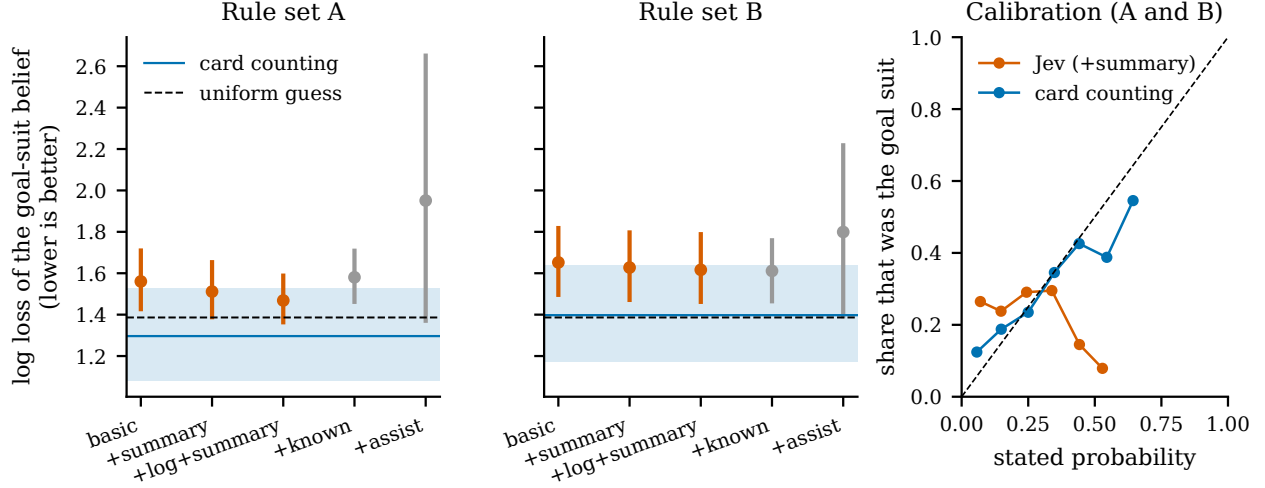


Figure 4: Jev’s belief is worse than card counting and worse than a uniform guess, and it is most wrong when it is most sure. Left and middle: log loss of the goal-suit belief (lower is better) for five state versions, 100 fundamentalist moments per rule set, 95% intervals. Grey: versions with numbers calculated by code (tests only). Blue line and band: card counting at the same moments. Right: calibration of the +summary version, both rule sets (800 suit probabilities per curve); a perfect belief lies on the diagonal.

intend. The behaviour text for the noise trader says “you take the current best bid as a loose anchor”. In B, Jev then buys with 57% of its probability; the twin buys with 8% (appendix Figure 8).

Neither wording is better. The algorithm wording gives the higher overlap for 5 of 6 types in A, but for only 2 of 6 in B. So which text works best depends on the market rules, and no single wording is a safe choice.

9 Jev earns less than its twin, and never clearly more

Figure 6 gives the answer to the question of the introduction, and Table 2 gives the numbers.

Jev loses to its twin. Over the 13 conditions of a rule set, Jev earns 33 chips per game less than its twin in A ($[-43, -24]$) and 58 less in B ($[-68, -47]$). For comparison, the fundamentalist twin earns 25 chips per game in A and 15 in B. No condition is clearly better than its twin. In A, 4 conditions are clearly worse and 9 are not clearly different. In B, 11 are clearly worse; only the two AS market-maker conditions are not.

The informed seat loses most. Jev with the fundamentalist persona loses 86 to 106 chips per game to the fundamentalist, in both rule sets and with both wordings. This agrees with section 7: the fundamentalist’s edge comes from its belief, and Jev’s belief is poor.

Close to the twin does not mean good. Jev is close to its twin where the twin does little. The market-maker twins earn little, and Jev as a market maker loses little. In these cases a small profit gap does not mean that Jev copies the twin: their decisions overlap only 0.04 to 0.10. Across the 24 persona conditions, a higher overlap in stage 1 goes with a *larger* profit gap in stage 2 (rank correlation between overlap and $|d|$: $+0.53$, permutation $p = 0.010$). The market makers cause most of this; inside one rule set, the correlation is also positive but not clear (A: $+0.43$, $p = 0.17$; B: $+0.50$, $p = 0.10$). So prediction H4b fails. At the levels of overlap that Jev reaches, the profit gap depends on which trader Jev plays, not on how well it copies that trader.

With no persona, Jev is worse than random trading in A. Neutral Jev earns less than the fundamentalist in the same seat and games (-119 in A, $[-153, -84]$; -81 in B, $[-117, -43]$). It also earns

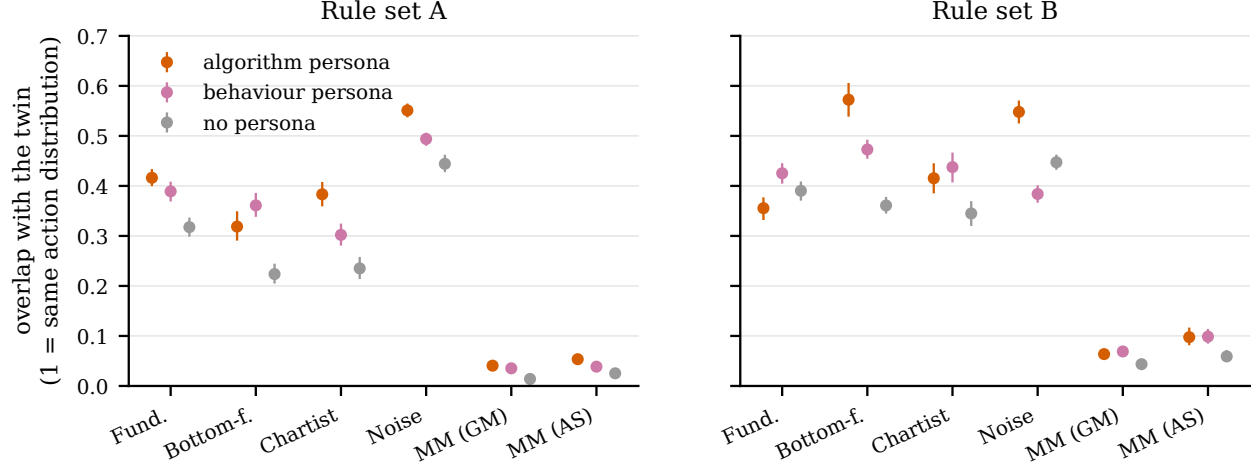


Figure 5: A persona moves Jev toward its twin, but not far. Overlap between Jev’s action probabilities and the twin’s at the same moment (1 = the same behaviour), for the algorithm persona, the behaviour persona and no persona. 100 moments per point, 95% intervals.

Condition	Rule set A (order book)			Rule set B (open outcry)		
	Jev	Twin	Jev – twin [95%]	Jev	Twin	Jev – twin [95%]
Neutral	–95	25	–119 [–153, –84]*	–65	15	–81 [–117, –43]*
Fund. (algorithm)	–81	25	–106 [–149, –63]*	–88	15	–103 [–150, –56]*
Fund. (behaviour)	–64	25	–89 [–124, –56]*	–70	15	–86 [–121, –50]*
Bottom-f. (algorithm)	–49	–9	–41 [–62, –19]*	–70	–9	–61 [–89, –33]*
Bottom-f. (behaviour)	–1	–9	+8 [–15, 32]	–59	–9	–51 [–77, –27]*
Chartist (algorithm)	–42	–34	–9 [–33, 15]	–76	–23	–52 [–80, –22]*
Chartist (behaviour)	–61	–34	–27 [–55, 1]	–81	–23	–57 [–91, –23]*
Noise (algorithm)	–45	–32	–13 [–41, 16]	–111	–35	–76 [–110, –40]*
Noise (behaviour)	–60	–32	–28 [–62, 5]	–112	–35	–77 [–105, –50]*
MM (GM) (algorithm)	–17	–7	–10 [–24, 6]	–58	–7	–51 [–70, –30]*
MM (GM) (behaviour)	–20	–7	–13 [–35, 7]	–58	–7	–51 [–74, –26]*
MM (AS) (algorithm)	–7	–11	+4 [–13, 22]	–46	–39	–7 [–38, 23]
MM (AS) (behaviour)	–3	–11	+8 [–10, 26]	–39	–39	+0 [–30, 29]

Table 2: Profit per game in the tested seat (chips), 40 paired games per cell. *: the 95% interval does not include 0. All 26 conditions use the same 40 seeds.

less than the noise trader, which decides at random, in A (–62, [–92, –34]). In B the difference to the noise trader is not clear (–30, [–67, 7]).

The market rules matter more than the words. Jev is closer to its twin in A than in B, by 25 chips per game ([12, 36]). The wording does not change profit: behaviour minus algorithm is +5 ([–5, 16]). The regression gives the same picture (Table 3): rule set A adds 20 chips ([6, 33]); the wording (0) and its interaction with the rule set (+11) have intervals that include 0. With the number of goal cards in the starting hand added, the other estimates do not change at all, because every condition plays the same deals; each extra goal card changes the gap by –5.8 chips ([–12.5, 1.7]), which is not clear.

How Jev loses. Jev trades about twice as often as its twin: 26 against 12 trades per game in its seat in A, and 19 against 9 in B. The whole market trades more when Jev is in it, and prices move away from true value. The mean distance between a trade price and the true card value (10 for a goal card, 0 for any other card) is 7.2 with Jev and 5.9 with the twin in A, and 9.5 against 6.2 in B (appendix Table 9). In B, the rules reject 17 of the twin’s orders per game, because they do not beat the best price; Jev has less than one

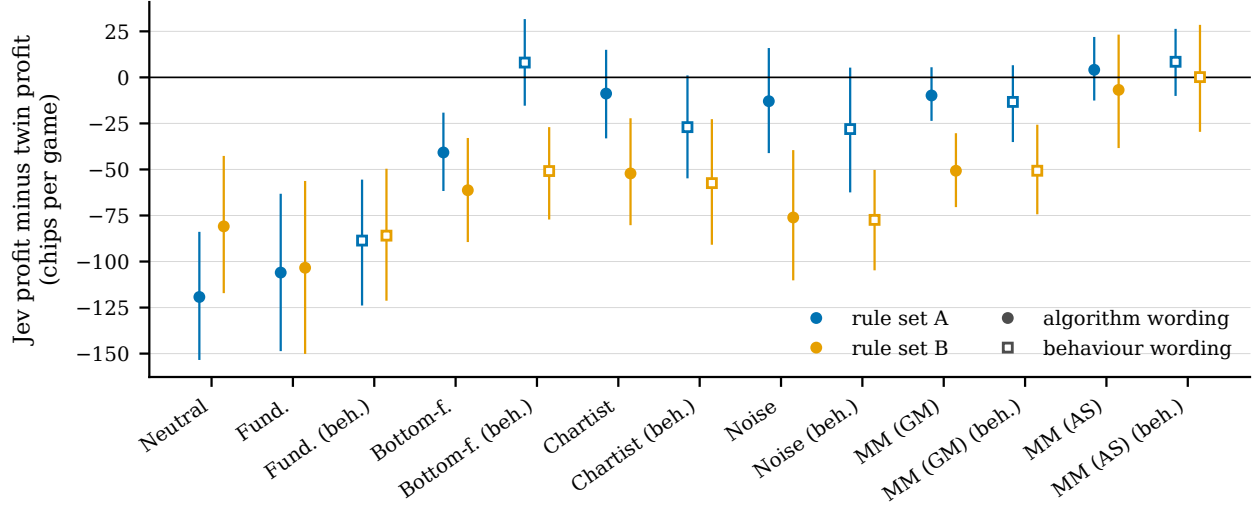


Figure 6: In full games, Jev earns less than the trader it plays, and never clearly more. Mean profit difference, Jev minus its twin, in the same seat and games (40 games per point, 95% intervals). “Neutral” is Jev with no persona; its twin is the fundamentalist. Filled points: algorithm persona; open squares: behaviour persona.

Term (reference: neutral Jev, rule set B, algorithm wording)	Estimate	95% interval
Intercept	−110	[−137, −83]
Rule set A	+20	[6, 33]
Behaviour wording	0	[−15, 15]
Rule set A × behaviour wording	+11	[−10, 29]
Type: fundamentalist	+2	[−29, 31]
Type: noise trader	+49	[13, 82]
Type: chartist	+61	[30, 92]
Type: bottom-feeder	+61	[36, 84]
Type: market maker (GM)	+66	[41, 90]
Type: market maker (AS)	+99	[65, 130]
Goal cards in the starting hand (second model)	−5.8	[−12.5, 1.7]

Table 3: The rule set changes the profit gap; the wording does not. Regression of d_g on the design, 1,040 pairs, bootstrap over games.

rejection per game, because its menu lists only accepted prices. So Jev does not lose because the market refuses it. It loses on the trades it makes.

The belief result holds in live games. Over all of Jev’s decisions in stage 2, the mean log loss of its goal-suit belief is 1.56 in A and 1.64 in B, against 1.28 and 1.30 for card counting at the same moments and 1.386 for a uniform guess. It stays flat from the first to the last 30 seconds (Figure 7).

10 Objections

We now answer the objections that we expect, one at a time.

1. “The personas are badly written.” Perhaps a better text would work. But we wrote each persona in two independent styles: one states the procedure, the other describes the behaviour. They fail in the same way, and the choice between them changes profit by +5 chips ([−5, 16]). The algorithm text for the fundamentalist contains the complete calculation. At the moment of Figure 1, this text did not stop Jev from selling the goal suit at 12 chips. A better text may exist, but the problem is not a missing instruction.

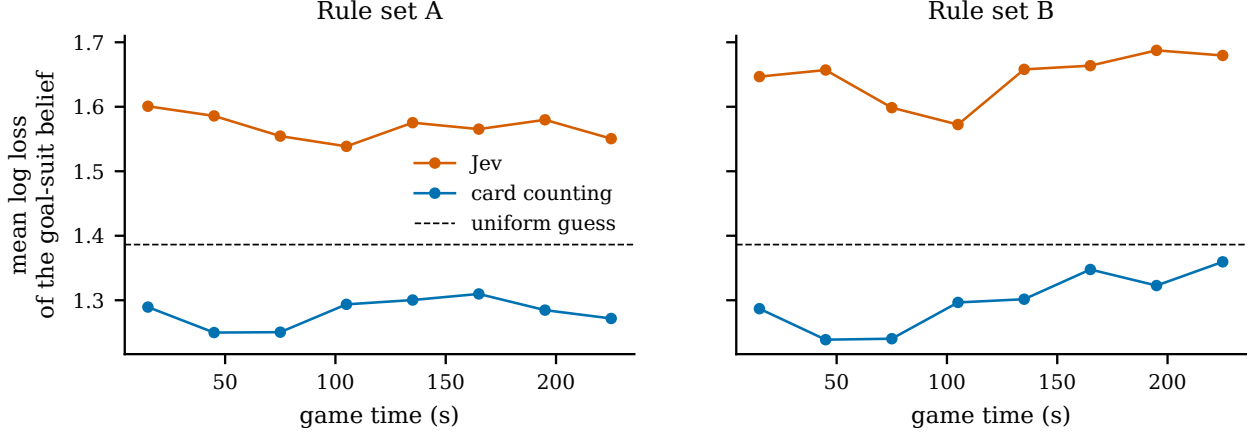


Figure 7: In full games, Jev’s belief does not improve as the game reveals cards. Log loss of the goal-suit belief in 30-second bins, all Jev decisions of a rule set, and card counting at the same moments. Dashed line: uniform guess.

2. “Jev did not have the information.” It did. The +log+summary version contains every trade, so it contains every fact that card counting uses. The +known version even gives the counts k_s directly. At the moment of Figure 1, that version made Jev more wrong (clubs 0.59). Only when we gave Jev the final probabilities (+assist) did it find the likely suit, and then it became overconfident.

3. “Sampling makes Jev random. Take its best option instead.” The twins are random by design: the fundamentalist draws its prices from a range. A trader that always takes one option cannot match them. And when an earlier version of this study took Jev’s most probable option, Jev passed at most of its turns (section 4). Sampling is the only rule that uses all of Jev’s answer.

4. “The twins are wrong, so the comparison is wrong.” Section 6 shows that the twins rank and vary as in the paper. Where our replication differs (10,000-event games, four noise traders), the difference does not reach stage 2, which uses 240-second games with a fundamentalist and a bottom-feeder in every game. One mismatch does reach the persona text: the bottom-feeder persona says that it copies “each other player”, while the twin copies only the fixed fundamentalist opponent, as in the paper. In this cell Jev’s text is wider than the twin’s rule.

5. “Card counting is a weak reference.” It is. It uses only the cards revealed by sales, and at the frozen moments its log loss is close to that of a uniform guess (1.30 in A and 1.40 in B, against 1.386). A stronger reference, for example one that models the other players’ behaviour, would make the gap to Jev larger, not smaller. Jev is already worse than a uniform guess.

6. “40 games are too few.” For one condition, they are: section 6 shows that 40 games detect only differences of about 28 to 43 chips. Smaller effects of a single persona can hide. But the main results pool 520 pairs per rule set, and their intervals are far from 0. The number of games was fixed before the run and is the same for every condition.

7. “This is about Figgie, not about markets.” Figgie is small, but it has the parts that matter: hidden information, a value that can be deduced, and other traders who react. We used two market rules, and the main result holds in both. If a model cannot play a trader here, where the rules are short and the algorithm is written out, we see no reason to expect more from it in a larger market.

8. “This tests one version of one model.” True. We tested `typesafe/jev-1.13`. The test itself does not depend on the model: any model that returns a choice from a menu can take Jev’s seat, and the twin arm, the seeds and the analysis stay the same.

11 Conclusion

We asked whether a decision model can be told which trader to be. For Jev in Figgie, the answer is: partly. Three measurements, each with its own data, give the same answer: Jev’s belief, its decisions at frozen moments, and its profit in full games. A persona changes what Jev does, and it does so reliably, but it does not give Jev the belief that makes a trader good. Jev follows the words of a persona and not its procedure. It reads the facts but does not draw the conclusion that the facts force. So it trades often, at poor prices, and on average it earns less than the trader it plays.

For market simulation, this means that a text persona can give a simulated participant a style, but not a judgement. A simulation that needs informed traders must build the information into the trader. It cannot ask for it in words, and, as the `+assist` test shows, it cannot simply hand the model the right numbers either.

Some questions stay open. We do not know whether Jev fails to read the facts or fails to use them: the `+known` test, which states the counts directly, suggests that reading is not the only problem. We do not know whether a model that gets the belief from a tool, and not as text in its state, would trade better. And we tested one model; other models may behave differently in the same seat.

It has not escaped our notice that the twin test applies to any model that is proposed as a simulated market participant.

Reproduction. All code, results and Jev request logs are in the repository, on the branch `claude/project-thread-vvzuh9`. Each output directory has a `run.json` with the code version and all settings. Each game record has the seed, the deal, every order, trade and cancel, and, for Jev seats, each decision with its probability and goal-suit belief. Each Jev request is logged with its game, seat and decision number.

```
python -m figgie.experiments.replicate --out results/final/stage0
S1=results/final/stage1
python -m figgie.experiments.stage1 moments --out $S1/moments.jsonl
for s in 1a 1b 1c; do
    python -m figgie.experiments.stage1 ask --study $s --backend jev \
        --moments $S1/moments.jsonl --out $S1/$s.jsonl
done
python -m figgie.experiments.stage1 analyse --moments $S1/moments.jsonl \
    --rows $S1/1a.jsonl $S1/1b.jsonl $S1/1c.jsonl --out $S1/report.json
python -m figgie.experiments.sweep --backend jev --games 40 --out results/final/stage2
python -m figgie.experiments.paired --run results/final/stage2 --stage1 $S1/report.json
python paper/final/make_figures.py
```

References

- [1] A. Ozerov, S. DiSilvio, Y. Luo. Traders in a strange land: agent-based discrete-event market simulation of the Figgie card game. arXiv:2110.00879, 2021.
- [2] Jane Street. Figgie. <https://www.janestreet.com/figgie/>.
- [3] TypeSafe AI. Jev decision model, `typesafe/jev-1.13`, via OpenRouter. 2026.
- [4] J. Horton. Large language models as simulated economic agents: what can we learn from homo silicus? NBER Working Paper 31122, 2023.
- [5] C. Chiarella, G. Iori, J. Perelló. The impact of heterogeneous trading rules on the limit order book and order flows. *J. Economic Dynamics and Control* 33(3):525–537, 2009.
- [6] L. Glosten, P. Milgrom. Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *J. Financial Economics* 14(1):71–100, 1985.
- [7] M. Avellaneda, S. Stoikov. High-frequency trading in a limit order book. *Quantitative Finance* 8(3):217–224, 2008.

A Tables and further figures

Setting	Line-up (4 of one type)	SD wealth	SD cash	SD payout	Trades per game
A-10k-paper	Noise	50.0	0.0	50.0	0
A-10k-paper	Fundamentalist	57.8	30.0	67.8	26
A-10k-paper	Bottom-feeder	50.0	0.0	50.0	0
A-240s-paper	Noise	50.0	0.0	50.0	0
A-240s-paper	Fundamentalist	57.4	27.8	66.8	25
A-240s-paper	Bottom-feeder	50.0	0.0	50.0	0
B-240s-paper	Noise	50.0	0.0	50.0	0
B-240s-paper	Fundamentalist	57.0	27.4	67.4	21
B-240s-paper	Bottom-feeder	50.0	0.0	50.0	0
A-240s-equal	Noise	50.0	0.0	50.0	0
A-240s-equal	Fundamentalist	57.9	25.9	65.5	22
A-240s-equal	Bottom-feeder	50.0	0.0	50.0	0
B-240s-equal	Noise	50.0	0.0	50.0	0
B-240s-equal	Fundamentalist	56.5	22.8	65.1	16
B-240s-equal	Bottom-feeder	50.0	0.0	50.0	0
Paper	Noise	201.0	188.5	61.4	–
Paper	Fundamentalist	61.8	33.1	68.6	–
Paper	Bottom-feeder	48.6	0	48.6	–

Table 4: Games of four traders of one type, as the paper’s table 2 (100 games each; standard deviations over games, averaged over seats).

Competitor	95% interval, slow minus fast (chips)		
	Wealth	Cash	Payout
slow0	[-8.2, 36.9]	[39.8, 54.1]	[-56.3, -8.9]
slow1	[-8.8, 33.3]	[43.5, 57.3]	[-60.2, -16.2]
slow2	[-11.4, 32.0]	[41.2, 53.9]	[-59.9, -15.0]
Paper, slow0	[15.7, 38.4]	[58.1, 65.8]	[-46.8, -23.0]

Table 5: Latency test (rule set A, 10,000 events, 100 games), as the paper’s table 3: one fast fundamentalist (delay 0) against three slow ones (delay 200, the paper’s 2Λ with $\Lambda = 100$).

Rule set	Type in the tested seat	Mean profit	SD profit	SD of d to fundamentalist
A	Fund.	39	85	–
A	Bottom-f.	-3	52	89
A	Chartist	-23	52	91
A	Noise	-38	35	97
A	MM (GM)	-11	32	80
A	MM (AS)	-11	48	96
B	Fund.	25	70	–
B	Bottom-f.	-4	47	76
B	Chartist	-18	66	82
B	Noise	-33	34	72
B	MM (GM)	-4	39	62
B	MM (AS)	-28	76	92

Table 6: Profit noise in the stage 2 line-up with rule-based traders only, equal speed, 100 games.

Rule set	State	Jev			Card counting	
		Log loss [95%]	Brier	Top-1	Log loss	Top-1
A	basic	1.560 [1.42, 1.72]	0.839	0.21	1.296	0.44
A	+summary	1.511 [1.38, 1.66]	0.814	0.22	1.296	0.44
A	+log+summary	1.468 [1.35, 1.60]	0.794	0.26	1.296	0.44
A	+known	1.580 [1.45, 1.72]	0.856	0.15	1.296	0.44
A	+assist	1.951 [1.36, 2.66]	0.857	0.45	1.296	0.44
B	basic	1.652 [1.49, 1.83]	0.891	0.14	1.397	0.41
B	+summary	1.627 [1.46, 1.81]	0.873	0.13	1.397	0.41
B	+log+summary	1.617 [1.45, 1.80]	0.861	0.15	1.397	0.41
B	+known	1.611 [1.45, 1.77]	0.876	0.13	1.397	0.41
B	+assist	1.799 [1.38, 2.23]	0.879	0.37	1.397	0.41

Table 7: Stage 1b: goal-suit belief, 100 fundamentalist moments per rule set. Uniform guess: log loss 1.386, Brier 0.75, top-1 0.25.

Rule set	Type	Overlap with twin			Change vs no persona	
		Algorithm	Behaviour	None	Algorithm	Behaviour
A	Fund.	0.42	0.39	0.32	+0.099*	+0.071*
A	Bottom-f.	0.32	0.36	0.22	+0.095*	+0.137*
A	Chartist	0.38	0.30	0.24	+0.148*	+0.067*
A	Noise	0.55	0.49	0.44	+0.107*	+0.050*
A	MM (GM)	0.04	0.04	0.01	+0.027*	+0.021*
A	MM (AS)	0.05	0.04	0.03	+0.028*	+0.013*
B	Fund.	0.36	0.43	0.39	-0.035*	+0.035*
B	Bottom-f.	0.57	0.47	0.36	+0.211*	+0.112*
B	Chartist	0.42	0.44	0.34	+0.070*	+0.093*
B	Noise	0.55	0.38	0.45	+0.101*	-0.063*
B	MM (GM)	0.06	0.07	0.04	+0.020*	+0.025*
B	MM (AS)	0.10	0.10	0.06	+0.038*	+0.040*

Table 8: Stage 1a: overlap with the twin, 100 moments per cell. *: Holm-corrected $p < 0.05$ (paired bootstrap over games).

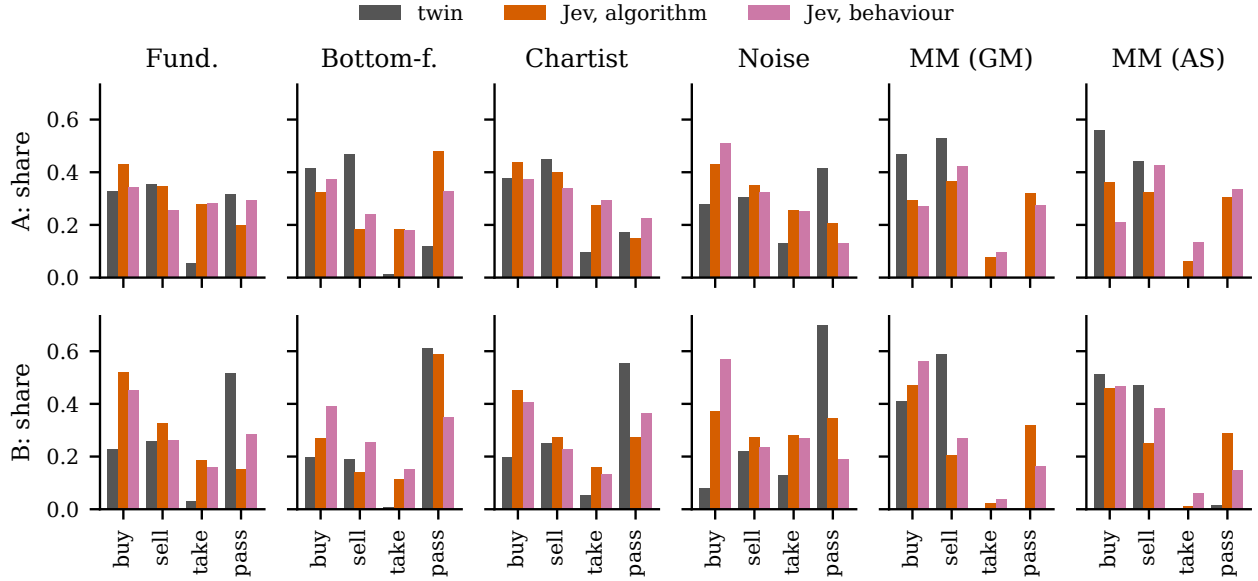


Figure 8: Stage 1a: share of probability on buying (bid or take the offer), selling, taking (either side) and passing.

Rule set	Condition	Seat trades		Passes		Mispricing		Belief log loss	
		Jev	Twin	Jev	Twin	Jev game	Twin game	Jev	Card c.
A	Neutral	34.8	16.0	41	28	8.31	5.97	1.61	1.10
B	Neutral	22.8	11.0	35	20	10.84	5.76	1.67	1.29
A	Fund.	33.6	16.0	27	28	9.45	5.97	1.70	1.20
B	Fund.	24.8	11.0	23	20	10.64	5.76	1.64	1.27
A	Fund. (beh.)	30.1	16.0	48	28	7.25	5.97	1.61	1.27
B	Fund. (beh.)	20.6	11.0	38	20	9.74	5.76	1.59	1.32
A	Bottom-f.	18.9	6.2	74	43	6.43	6.04	1.58	1.28
B	Bottom-f.	11.8	3.2	83	54	9.15	6.31	1.60	1.31
A	Bottom-f. (beh.)	23.0	6.2	47	43	6.62	6.04	1.50	1.34
B	Bottom-f. (beh.)	16.5	3.2	50	54	8.23	6.31	1.66	1.18
A	Chartist	19.7	6.0	57	80	6.62	6.36	1.55	1.33
B	Chartist	12.9	1.9	73	100	9.03	6.99	1.56	1.29
A	Chartist (beh.)	24.8	6.0	49	80	9.42	6.36	1.65	1.42
B	Chartist (beh.)	17.3	1.9	57	100	12.43	6.99	1.73	1.43
A	Noise	32.8	24.3	25	47	7.62	5.93	1.47	1.42
B	Noise	25.5	11.4	54	69	9.75	6.35	1.69	1.36
A	Noise (beh.)	36.9	24.3	20	47	7.94	5.93	1.58	1.28
B	Noise (beh.)	26.5	11.4	32	69	9.63	6.35	1.65	1.31
A	MM (GM)	20.2	8.1	42	66	5.78	5.19	1.53	1.23
B	MM (GM)	15.2	12.9	37	44	9.17	4.76	1.62	1.27
A	MM (GM) (beh.)	23.1	8.1	41	66	6.01	5.19	1.52	1.25
B	MM (GM) (beh.)	17.5	12.9	23	44	8.66	4.76	1.69	1.30
A	MM (AS)	20.1	9.7	37	32	6.55	5.96	1.51	1.24
B	MM (AS)	13.5	11.1	34	38	7.71	6.92	1.65	1.28
A	MM (AS) (beh.)	21.9	9.7	42	32	5.93	5.96	1.50	1.22
B	MM (AS) (beh.)	18.6	11.1	24	38	8.26	6.92	1.56	1.23

Table 9: Stage 2, secondary measures (means per game, 40 games per cell). Seat trades: trades of the tested seat. Passes: decisions with no order. Mispricing: mean $|\text{price} - \text{value}|$ over all trades of the game, with value 10 for a goal card and 0 for other cards. Belief: mean log loss of Jev’s goal-suit belief over its decisions, and of card counting at the same moments (uniform guess: 1.386).